

EXPERIMENT – 13

CODE:

```
#include <reg51.h>

void main(void)
{
    unsigned int i;
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
    unsigned long sum = 0;

    for(i = 0; i < 5; i++)
    {
        sum = sum + array[i];
    }

    P0 = (unsigned char)(sum & 0xFF);    // Lower 8 bits
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits

    while(1);
}
```

OUTPUT:

The screenshot displays the Keil uVision IDE interface for a project named "LAB - 13.uvproj". The main window shows the assembly view of the program, with the following code:

```
1: void main(void)
2: {
3:     unsigned int i;
4:     unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
5:     unsigned long sum = 0;
6:     for(i = 0; i < 5; i++)
7:     {
8:         sum = sum + array[i];
9:     }
10:    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
11:    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
12:    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
13:    while(1);
14: }
```

The registers window on the left shows the current state of the registers, with the PC register at address 0x0909. The command window at the bottom shows the command "Load "E:\\SINATS COURSES\\EMBEDDED SYSTEMS\\EMBEDDED C\\Objects\\LAB - 13" and the simulation status bar at the bottom indicates the simulation is running at 11:136.99402500 sec.

Three parallel port windows are open, showing the output of the program:

- Parallel Port 2: P2: 0x01, Pins: 0x01
- Parallel Port 1: P1: 0xAB, Pins: 0xAB
- Parallel Port 0: P0: 0xCC, Pins: 0xCC

The status bar at the bottom of the IDE shows the simulation time, the current location (L6 C1), and the status of the simulation (CAP, NUM, SCRL, OVR, R/W).